Lab 7: Fuzzing for Input Validation Bugs

Name: Sharat Doddihal SRN: PES1UG24CS430
Repo: L6_L7_07_hospital_traige_priority (folder: Lab 7)
Module under test: processor.py - parse_vitals_summary, compute_patient_bmi, assign_ward_bed

Step 2: Run the buggy application

Command: python processor.py
Valid input works, but an incomplete vitals string (98,95) crashes with IndexError.

Step 2 - python processor.py (buggy)

```
$ python processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): 98,95,37.2
Summary: Vitals: HR=98.0, SpO2=95.0% - STABLE

==== Running compute_patient_bmi ====
Enter weight/height (e.g. 70/1.75): 70/1.75
BMI: 22.86

==== Running assign_ward_bed ====
Enter allocation (e.g. ICU-05): ICU-05
Assignment: Intensive Care Unit Bed #5

$ python processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): 98,95
Traceback (most recent call last):
  File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/processor.py", line 41, in <module>
    main()
  File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/processor.py", line 30, in main
    print("Summary:", parse_vitals_summary(v_input))
  File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/processor.py", line 5, in parse_vitals_summary
    temp = float(parts[2])
IndexError: list index out of range
```

Step 4: Run the Hypothesis fuzz tests against the buggy code

Command: python -m pytest test_processor.py

All 3 tests fail. Hypothesis found 9 distinct crashes (IndexError, ValueError, AttributeError) on inputs such as '', '0', '/', '-' and None.

Step 4 - pytest test_processor.py (before fix)

```
$ python -m pytest test_processor.py
============================= test session starts ==============================
platform linux -- Python 3.10.12, pytest-9.2.0, pluggy-1.7.0
rootdir: /sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy
plugins: hypothesis-6.168.5
collected 3 items

test_processor.py FFF                                                    [100%]

=================================== FAILURES ===================================
_____________________ test_parse_vitals_summary_no_crash ______________________
  + Exception Group Traceback (most recent call last):
  |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 11, in test_parse_vitals_summary_no_crash
  |     @given(st.text() | st.none())
  | exceptiongroup.BaseExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 14, in test_parse_vitals_summary_no_crash
    |     result = parse_vitals_summary(s)
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/processor.py", line 4, in parse_vitals_summary
    |     spo2 = float(parts[1])
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 16, in test_parse_vitals_summary_no_crash
    |     pytest.fail(f"parse_vitals_summary crashed on {s!r}: {type(e).__name__}: {e}")
    | Failed: parse_vitals_summary crashed on '0': IndexError: list index out of range
    | Failing test case: test_parse_vitals_summary_no_crash(
    |     s='0',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 14, in test_parse_vitals_summary_no_crash
    |     result = parse_vitals_summary(s)
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/processor.py", line 3, in parse_vitals_summary
    |     hr = float(parts[0])
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 16, in test_parse_vitals_summary_no_crash
    |     pytest.fail(f"parse_vitals_summary crashed on {s!r}: {type(e).__name__}: {e}")
    | Failed: parse_vitals_summary crashed on '': ValueError: could not convert string to float: ''
    | Failing test case: test_parse_vitals_summary_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 14, in test_parse_vitals_summary_no_crash
    |     result = parse_vitals_summary(s)
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/processor.py", line 2, in parse_vitals_summary
    |     parts = vitals_csv.split(',')
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 16, in test_parse_vitals_summary_no_crash
    |     pytest.fail(f"parse_vitals_summary crashed on {s!r}: {type(e).__name__}: {e}")
    | Failed: parse_vitals_summary crashed on None: AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_parse_vitals_summary_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________ test_compute_patient_bmi_safe ______________________
  + Exception Group Traceback (most recent call last):
  |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 21, in test_compute_patient_bmi_safe
  |     @given(st.text() | st.none())
  | exceptiongroup.BaseExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 24, in test_compute_patient_bmi_safe
    |     result = compute_patient_bmi(s)
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/processor.py", line 11, in compute_patient_bmi
    |     weight = float(w_str)
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
```

```
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 26, in test_compute_patient_bmi_safe
    |     pytest.fail(f"compute_patient_bmi crashed on {s!r}: {type(e).__name__}: {e}")
    | Failed: compute_patient_bmi crashed on '/': ValueError: could not convert string to float: ''
    | Failing test case: test_compute_patient_bmi_safe(
    |     s='/',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 24, in test_compute_patient_bmi_safe
    |     result = compute_patient_bmi(s)
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/processor.py", line 10, in compute_patient_bmi
    |     w_str, h_str = bmi_string.split('/')
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 26, in test_compute_patient_bmi_safe
    |     pytest.fail(f"compute_patient_bmi crashed on {s!r}: {type(e).__name__}: {e}")
    | Failed: compute_patient_bmi crashed on '': ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_compute_patient_bmi_safe(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 24, in test_compute_patient_bmi_safe
    |     result = compute_patient_bmi(s)
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/processor.py", line 10, in compute_patient_bmi
    |     w_str, h_str = bmi_string.split('/')
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 26, in test_compute_patient_bmi_safe
    |     pytest.fail(f"compute_patient_bmi crashed on {s!r}: {type(e).__name__}: {e}")
    | Failed: compute_patient_bmi crashed on None: AttributeError: 'NoneType' object has no attribute 'split'
    | Failing test case: test_compute_patient_bmi_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________ test_assign_ward_bed_safe _____________________
  + Exception Group Traceback (most recent call last):
  |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 31, in test_assign_ward_bed_safe
  |     @given(st.text() | st.none())
  | exceptiongroup.BaseExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 34, in test_assign_ward_bed_safe
    |     result = assign_ward_bed(s)
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/processor.py", line 24, in assign_ward_bed
    |     bed_num = int(bed_str)
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 36, in test_assign_ward_bed_safe
    |     pytest.fail(f"assign_ward_bed crashed on {s!r}: {type(e).__name__}: {e}")
    | Failed: assign_ward_bed crashed on '-': ValueError: invalid literal for int() with base 10: ''
    | Failing test case: test_assign_ward_bed_safe(
    |     s='-',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 34, in test_assign_ward_bed_safe
    |     result = assign_ward_bed(s)
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/processor.py", line 17, in assign_ward_bed
    |     ward, bed_str = allocation_code.strip().split('-')
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 36, in test_assign_ward_bed_safe
    |     pytest.fail(f"assign_ward_bed crashed on {s!r}: {type(e).__name__}: {e}")
    | Failed: assign_ward_bed crashed on '': ValueError: not enough values to unpack (expected 2, got 1)
    | Failing test case: test_assign_ward_bed_safe(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 34, in test_assign_ward_bed_safe
    |     result = assign_ward_bed(s)
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/processor.py", line 17, in assign_ward_bed
    |     ward, bed_str = allocation_code.strip().split('-')
    | AttributeError: 'NoneType' object has no attribute 'strip'
    |
    | During handling of the above exception, another exception occurred:
```

```
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/sessions/rcw-01unr6djybkujuxqtzez3lrq/lab7run/buggy/test_processor.py", line 36, in test_assign_ward_bed_safe
    |     pytest.fail(f"assign_ward_bed crashed on {s!r}: {type(e).__name__}: {e}")
    | Failed: assign_ward_bed crashed on None: AttributeError: 'NoneType' object has no attribute 'strip'
    | Failing test case: test_assign_ward_bed_safe(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
========================== short test summary info ===========================
FAILED test_processor.py::test_parse_vitals_summary_no_crash - exceptiongroup...
FAILED test_processor.py::test_compute_patient_bmi_safe - exceptiongroup.Base...
FAILED test_processor.py::test_assign_ward_bed_safe - exceptiongroup.BaseExce...
============================ 3 failed in 9.62s ============================
```

Step 5: Run the fixed application

Command: python processor.py

Valid input gives the same results; malformed, non-numeric, NaN/inf, zero-height and zero-bed inputs now return error messages instead of crashing.

Step 5 - python processor.py (fixed)

```
$ python processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): 98,95,37.2
Summary: Vitals: HR=98.0, SpO2=95.0% - STABLE

==== Running compute_patient_bmi ====
Enter weight/height (e.g. 70/1.75): 70/1.75
BMI: 22.86

==== Running assign_ward_bed ====
Enter allocation (e.g. ICU-05): ICU-05
Assignment: Intensive Care Unit Bed #5

$ python processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): 98,95
Summary: Invalid input: expected HR,SpO2,Temp

==== Running compute_patient_bmi ====
Enter weight/height (e.g. 70/1.75): /
BMI: Invalid input: expected weight/height

==== Running assign_ward_bed ====
Enter allocation (e.g. ICU-05): -
Assignment: Invalid input: unknown ward ''

$ python processor.py
==== Running parse_vitals_summary ====
Enter vitals (HR,SpO2,Temp): abc,nan,inf
Summary: Invalid input: vitals must be numbers

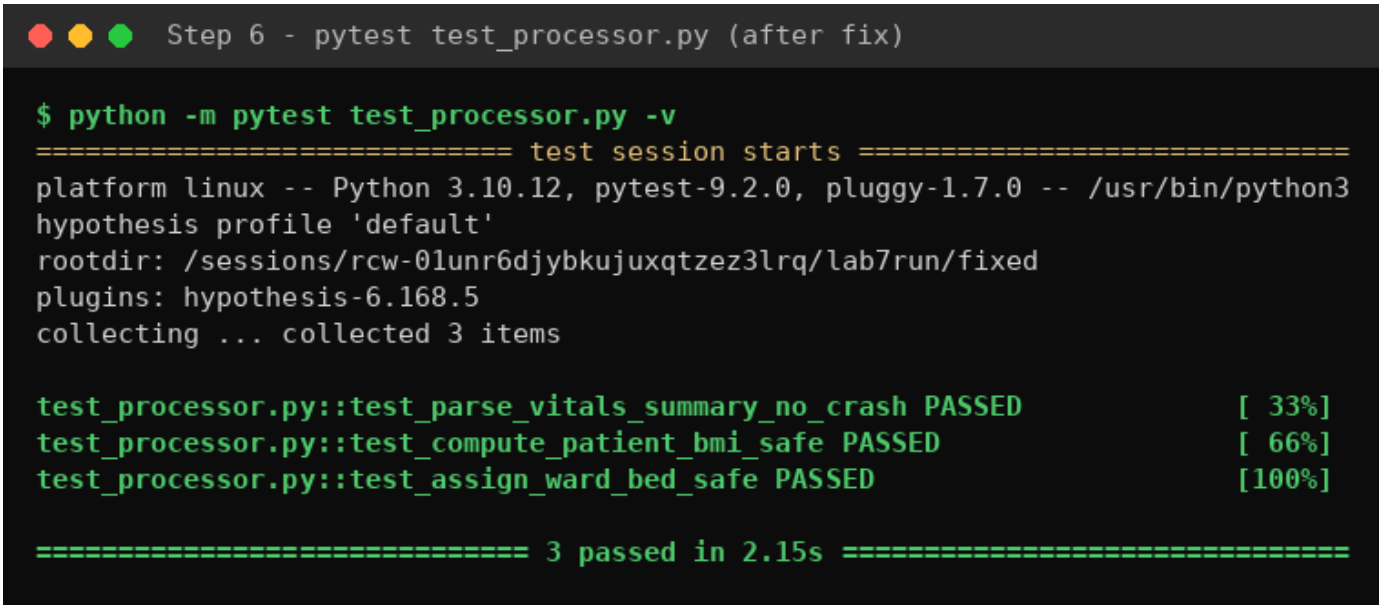
==== Running compute_patient_bmi ====
Enter weight/height (e.g. 70/1.75): 70/0
BMI: Invalid input: expected weight/height

==== Running assign_ward_bed ====
Enter allocation (e.g. ICU-05): ICU-0
Assignment: Invalid input: bed number must be positive
```

Step 6: Re-run the fuzz tests against the fixed code

Command: python -m pytest test_processor.py -v

All 3 tests pass (500 generated examples each).



Fixes made in processor.py

parse_vitals_summary: rejects non-strings, requires exactly 3 comma-separated fields, and parses each with _parse_number(), which rejects non-numeric, NaN and infinite values.
compute_patient_bmi: rejects non-strings and anything that isn't exactly weight/height; weight and height must be finite and positive; overflow and non-finite results return None.
assign_ward_bed: rejects non-strings and anything that isn't exactly WARD-BED; unknown wards and non-ASCII-digit or non-positive bed numbers return an error message (no KeyError).
main(): prints a readable message when compute_patient_bmi returns None.